

Telecom Portfolio Risk Report

Generated 13 August 2026 · portfolio state, not a time series

Unmarked	OBSERVED - measured from the source data.
†	PROJECTED - derived from the stated economic assumptions, not measured.
‡	SIMULATED - from generated history. Shows portfolio structure only; churn in it is flat by construction and cannot support a forecast.

Portfolio at a glance

3,333	14.49%	483
Customers	Churn rate	Churned
\$198,146	\$59.45	\$31,567
Revenue	ARPU	Revenue at risk

The most decision-relevant fact is that the portfolio has a churn rate of 14.49%, resulting in revenue at risk of 31566.93. The confirmed churn drivers are 4+ customer service calls, international plan subscription, and heavy daytime usage, defined as a day charge of 45 or more. The customer group that needs attention first is the 267 high service call customers, as they are at a higher risk of churn due to exceeding the 4+ service call threshold. Additionally, the 323 international plan subscribers and 210 heavy day usage customers also require attention due to their increased churn risk. The total revenue of 198146.03 is at risk due to these factors, with the international plan subscribers and heavy day usage customers contributing to this risk. The high churn rate and significant revenue at risk necessitate immediate attention to these customer groups.

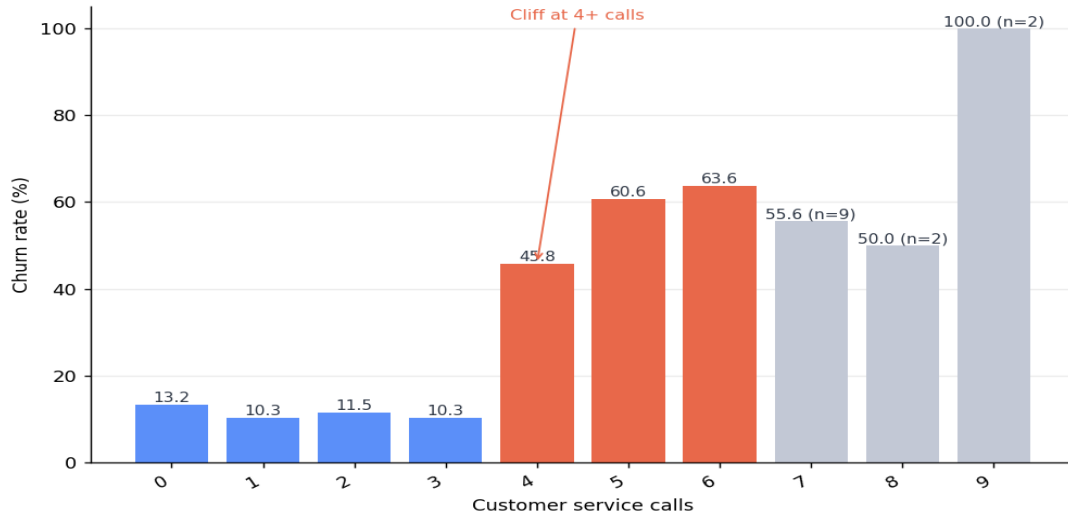
Confirmed churn drivers

We've identified three confirmed drivers of churn: international plan subscription, 4 or more customer service calls, and heavy daytime usage, defined as a day charge of 45 or more. Each of these drivers represents a threshold, rather than a gradual trend, which is important for targeting. For example, having 4 or more customer service calls raises the churn rate from 11.25% to 51.69%. Similarly, subscribing to an international plan results in a churn rate of 43.72% or 39.13%, compared to 13.85% or 5.3% for non-international plans. Heavy daytime usage also significantly increases churn, from 5.0% to 59.0% among customers with no other risk factors. These thresholds suggest that targeting customers who have crossed these thresholds can be an effective way to address churn.

Driver	Threshold	Below	Above	Customers
Customer service calls	4 or more	11.25%	51.69%	267
International plan	subscribed	11.5%	42.41%	323
Daytime charge	\$45 or more	11.43%	60.0%	210

Churn by number of customer service calls

Churn is flat from 0 to 3 calls, then jumps sharply. This is a cliff, not a slope.

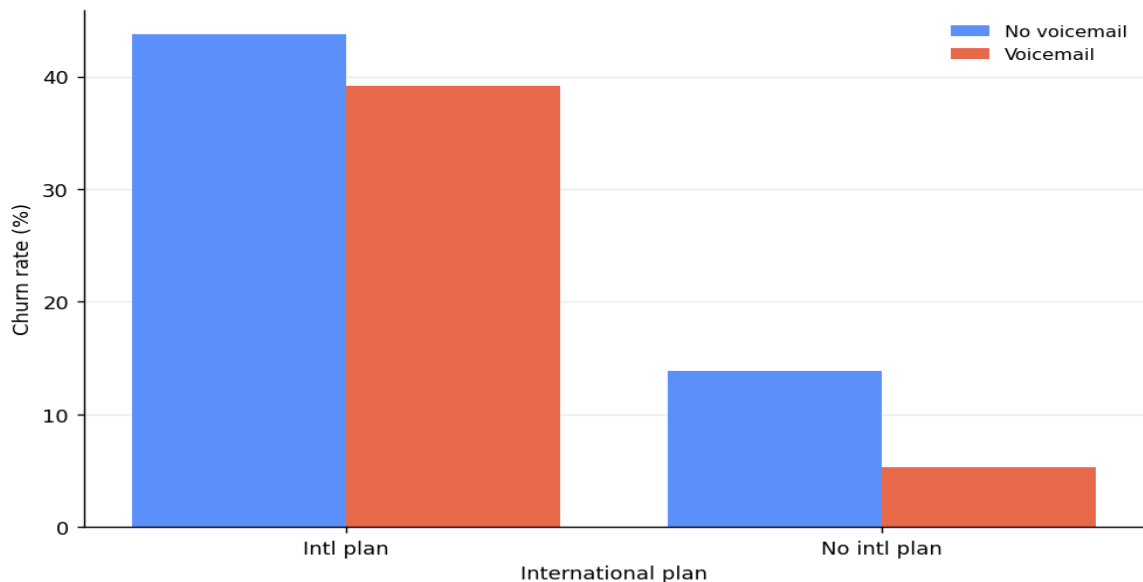


Bucket as '0-3' vs '4+'. Treating call count as a continuous predictor understates the effect badly. Greyed bars have fewer than 20 customers and are too small to read as evidence.

AI explanation. The pattern shows that the churn rate remains roughly the same for customers with 0, 1, 2, and 3 service calls, with rates of 13.2%, 10.33%, 11.46%, and 10.26% respectively. However, at the fourth call, the churn rate jumps sharply to 45.78%. This suggests that a fourth call may imply that the first three calls did not resolve the issue, leading to increased frustration and a higher likelihood of churn. To address this, the company should review and improve its issue resolution process for customers who have made three or more service calls, such as by assigning a dedicated support specialist to ensure that issues are fully resolved.

Churn by plan combination

International plan holders churn several times higher.

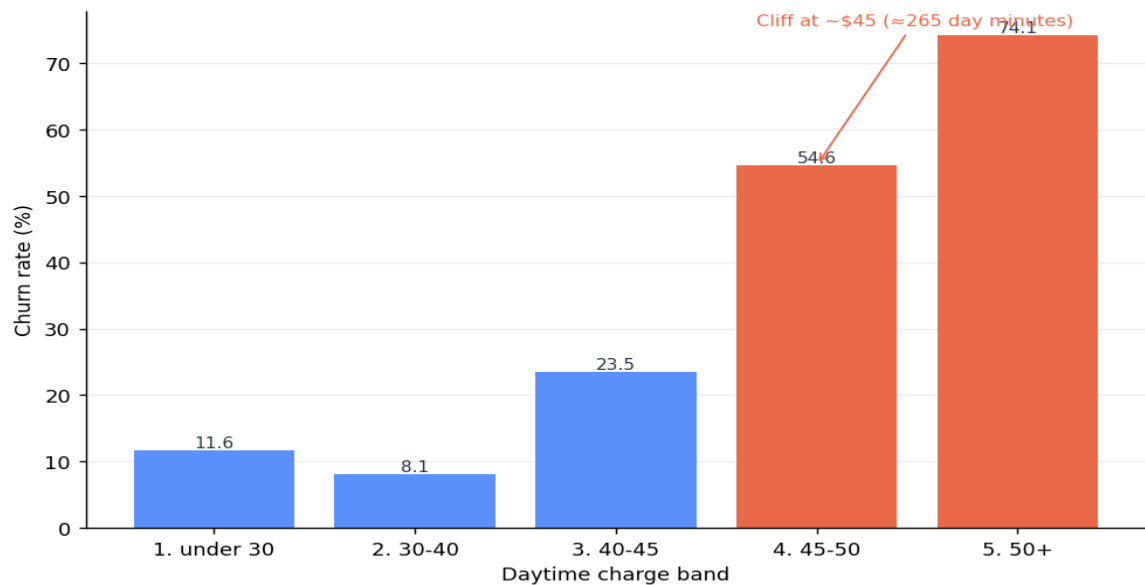


Voicemail plan has little independent effect; the international plan is the driver.

AI explanation. The pattern shows a sharp change in churn rate when comparing international plan holders to those without, with churn rates of 43.72% and 39.13% for international plan holders versus 13.85% and 5.3% for non-international plan holders. This suggests that the international plan may be a key factor in driving churn, possibly due to poor value for typical usage or higher price sensitivity among these customers. In contrast, the presence or absence of a voicemail plan appears to have a relatively minor impact on churn rates. To address this issue, the company should review and adjust the pricing and benefits of the international plan to better align with customer needs and expectations, such as by offering more competitive rates or additional features to justify the cost.

Churn by daytime charge band

Churn DIPS before it explodes. A linear model of day charge would find almost nothing.



Found while building the Phase 6 rules engine. The earlier two-driver segmentation reported a 'baseline' of 8.2% that concealed this group.

AI explanation. The churn rate drops from 11.64% in the under 30 daytime charge band to 8.12% in the 30-40 band, before rising steeply to 23.53% in the 40-45 band and further to 54.61% and 74.14% in the higher bands. This non-monotonic pattern suggests that customers may be experiencing bill shock or are on the wrong plan for their usage, leading to high churn rates. The significant increase in churn rate from the 30-40 band to the 40-45 band, where the average daytime minutes are 247.7, indicates a potential mismatch between customer usage and their current plan. To address this, the company should review the plans of customers in the higher daytime charge bands, such as the 152 customers in the 45-50 band who generate \$11613.29 in revenue, and consider offering more tailored plans to reduce churn.

Risk segments and recommended actions

The data shows that the Critical segment, with all three risk drivers, has a churn rate of 100.0% and revenue at risk of 81.98. The driver of this high churn rate is the combination of 4+ customer service calls, international plan subscription, and heavy daytime usage. To address this, a senior retention call with a premium support offer and 20% discount for 3 periods is recommended, targeting 1 customer. The data supports this action, with a projected net value of 241.6, assuming a save rate of 0.3. The next priority is the Severe segment with 4+ service calls and a second driver, which has a churn rate of 66.67% and revenue at risk of 1834.15. The driver of this high churn rate is the combination of service calls and a second risk driver. To address this, a senior retention specialist call with a premium support tier and 15% discount for 3 periods is recommended, targeting 42 customers. The data supports this action, with a projected net value of 5752.2, assuming a save rate of 0.3. These segments are ranked by urgency, with the Critical segment being the highest priority due to its extremely high churn rate, followed by the Severe segments. The projected values depend on the stated assumptions and are not measurements.

Segment	Customers	Churn	Revenue at risk	Net value†
Critical — all three risk drivers	1	100.0%	\$82	\$242†
Severe — 4+ service calls with a second driver	42	66.67%	\$1,834	\$5,752†
Severe — heavy daytime usage	194	59.28%	\$9,226	\$29,017†
High — 4+ service calls only	224	48.66%	\$5,549	\$20,044†
Elevated — international plan only	267	37.83%	\$6,139	\$20,823†
Baseline — no risk drivers	2,605	4.95%	\$8,738	-

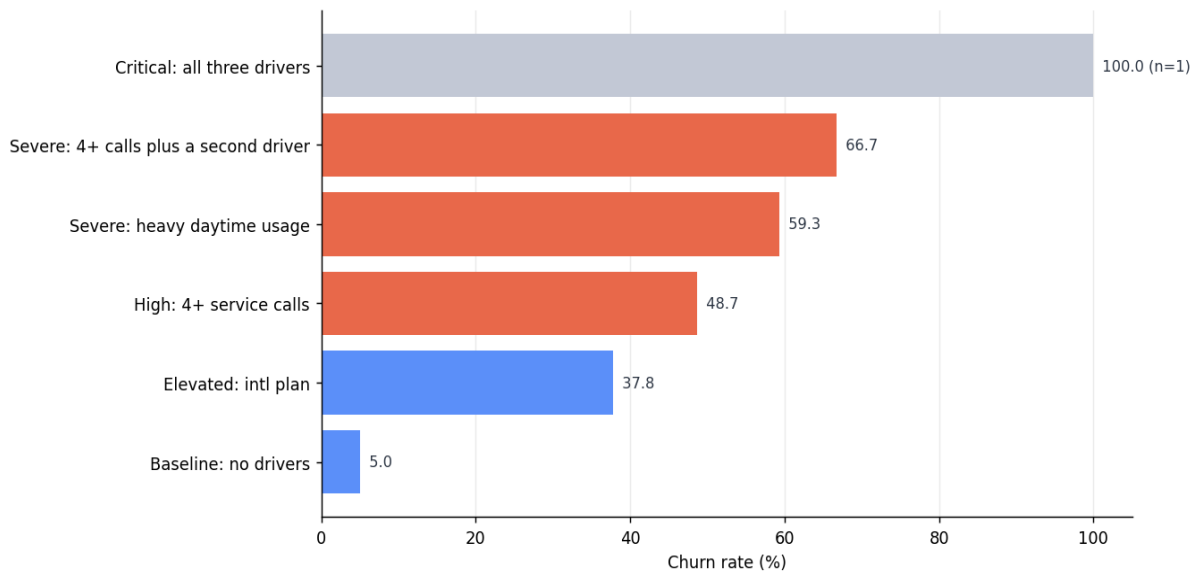
Recommended actions

Segment	Action	Offer	Customers
Critical — all three risk drivers	Immediate senior retention call, same day	Premium support + tariff review + 20% discount, 3 periods	1
Severe — 4+ service calls with a second driver	Senior retention specialist call within 48 hours	Premium support tier + 15% discount for 3 periods	42
Severe — heavy daytime usage	Tariff review call within 5 working days	Move to a daytime bundle or higher-allowance plan	194
High — 4+ service calls only	Proactive support review call within 5 working days	Service credit + dedicated support contact	224
Elevated — international plan only	Plan review — check the tariff fits actual usage	Tariff optimisation or downgrade without penalty	267

† **Assumptions behind projected values:** save rate 30% (band 15%-45%), contact cost \$8.00, acquisition cost \$45.00, horizon 12 periods. These are industry-typical placeholders, not measurements: the dataset contains no cost data.

Churn rate by risk segment

Segments built from the three confirmed drivers.

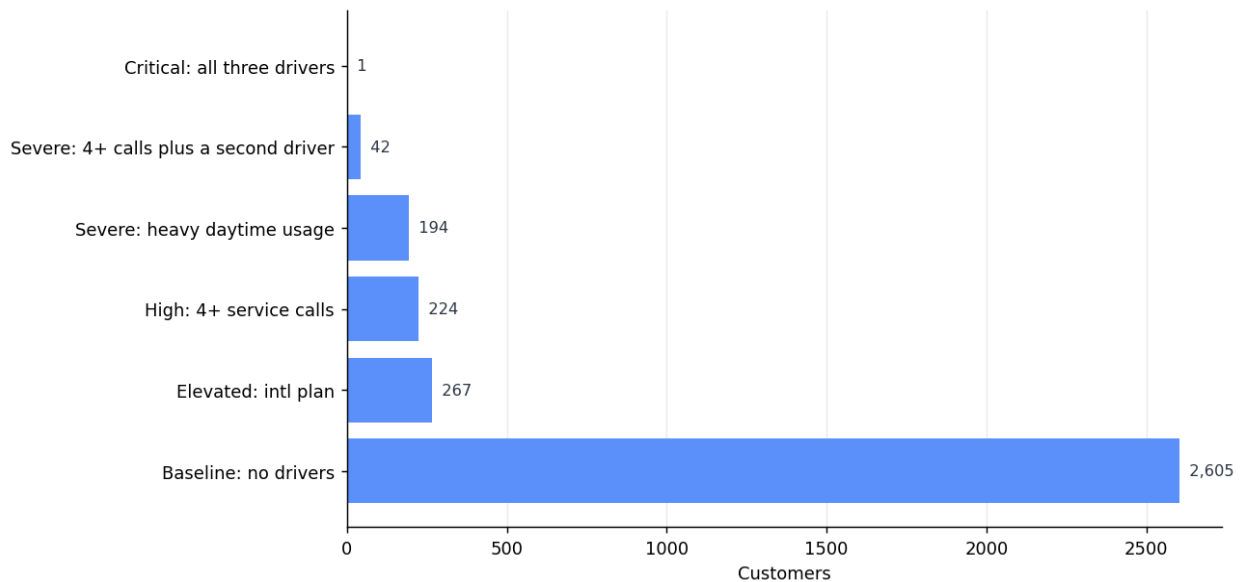


Segments are mutually exclusive and exhaustive: counts sum to 3,333 and revenue to the portfolio total. Greyed bars have fewer than 20 customers and are too small to read as evidence.

AI explanation. The pattern in the chart shows a sharp change in churn rate at the highest severity ranks, with the "Critical: all three drivers" segment having a churn rate of 100.0% and the "Severe: 4+ calls plus a second driver" segment having a churn rate of 66.67%. The "Severe: heavy daytime usage" segment also has a high churn rate of 59.28%. Given the size and churn rate of these segments, the "Severe: heavy daytime usage" segment deserves attention first, with 194 customers and revenue at risk of 9225.97. To address this, a specific step would be to develop a targeted retention program for customers with heavy daytime usage, focusing on the 115 customers who have already churned in this segment.

Customers per risk segment

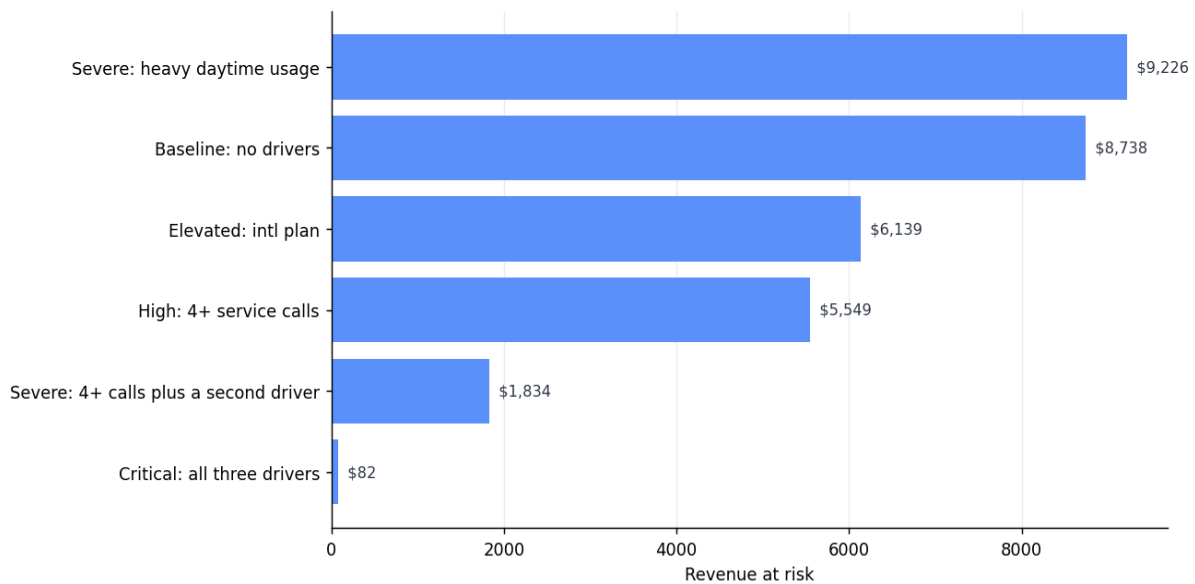
Most of the base carries no risk driver at all.



Segment sizes are highly uneven; read rates alongside n.

Revenue at risk by segment

Revenue attached to customers who churned, by segment. The largest exposure is not the highest-rate segment.



Rate and exposure rank differently. A large low-rate segment can carry more absolute revenue risk than a small high-rate one.

AI explanation (UNVALIDATED). The pattern shows that the Baseline segment has the largest revenue at risk, \$8737.56, despite not having the highest churn rate, which is 4.95%. This suggests that a low rate applied to a very large base, such as the 2605 customers in the Baseline segment, can still produce substantial absolute loss. The Baseline segment carries the most absolute revenue risk because it has the largest number of customers, resulting in a higher total revenue at risk. To address this, the company should prioritize retention efforts in the Baseline segment, such as offering targeted promotions or improved customer service, to reduce the \$8737.56 revenue at risk.

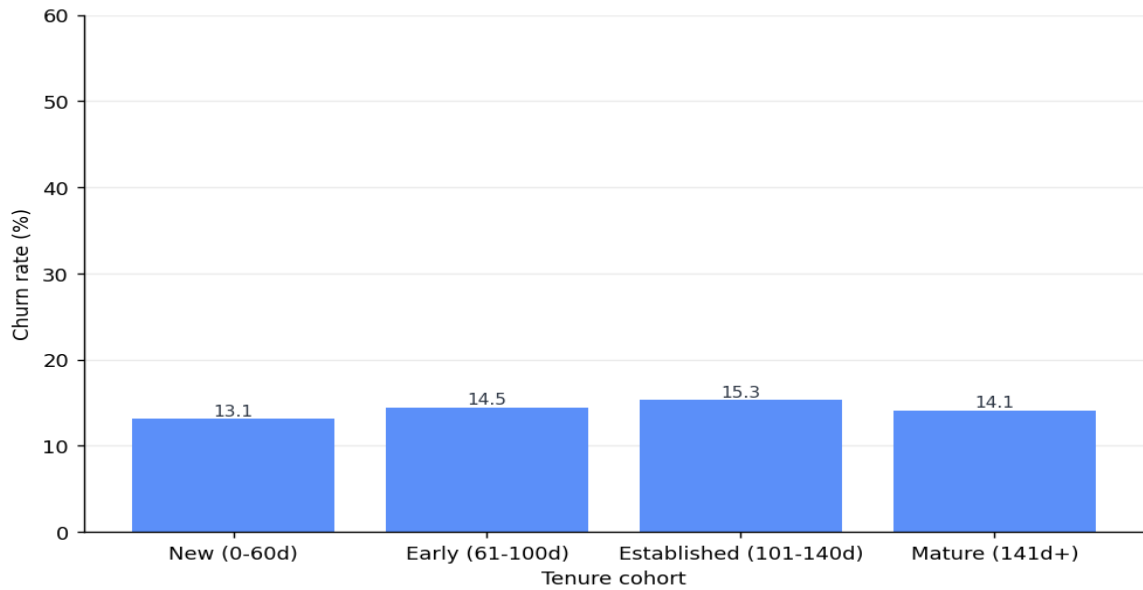
Tenure: an expected driver that is not one

The data shows that churn rates are close together across all tenure cohorts, with rates of 13.11%, 14.48%, 15.29%, and 14.12% for the New, Early, Established, and Mature cohorts, respectively. This suggests that tenure is not a significant predictor of churn, which is useful to know because it means that retention efforts should be targeted based on customer behavior rather than how long someone has been a customer. The confirmed drivers of churn, such as 4+ customer service calls, international plan subscription, and day_charge >= 45, should be the focus of retention efforts. Therefore, it is recommended that retention efforts be targeted towards customers with these specific behaviors, rather than solely based on their tenure.

Cohort	Customers	Churn	ARPU	Avg service calls
New (0-60d)	511	13.11%	\$59.70	1.52
Early (61-100d)	1,153	14.48%	\$59.13	1.57
Established (101-140d)	1,138	15.29%	\$59.49	1.6
Mature (141d+)	531	14.12%	\$59.83	1.5

Portfolio structure by tenure cohort

Churn is flat across cohorts: 13.1% to 15.3%.

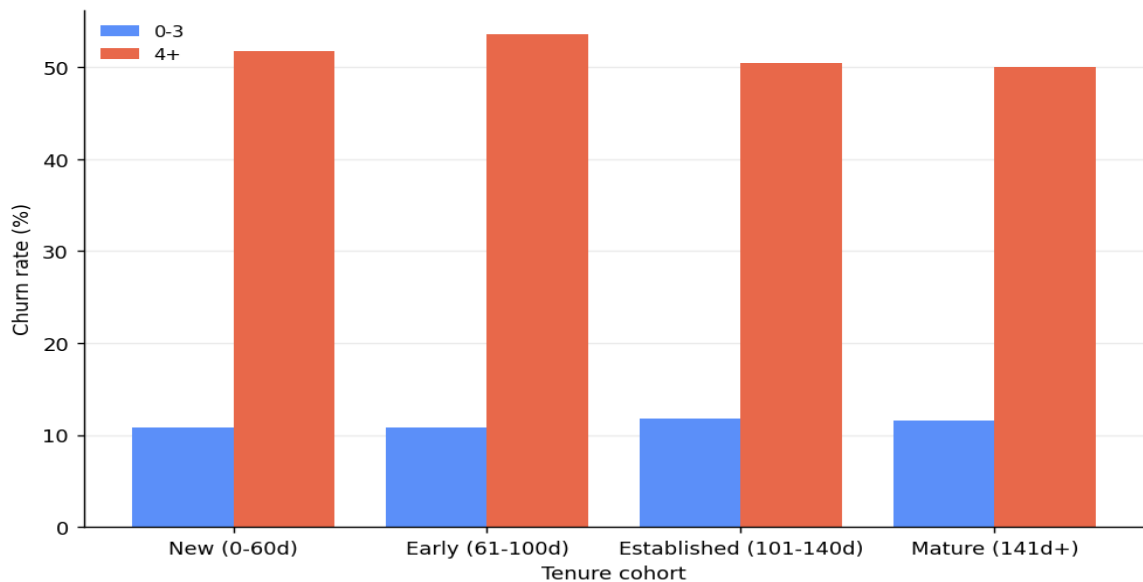


Plotted on the same y-scale as the driver charts to show how little variation there is here by comparison.

AI explanation. The pattern of churn rates across tenure cohorts is flat, ranging from 13.1% to 15.3%, with no significant variation between cohorts. This suggests that tenure does not predict churn, and targeting based on tenure may not be effective. The small spread between cohorts, with the highest churn rate of 15.29% in the Established cohort and the lowest of 13.11% in the New cohort, is not a meaningful pattern. To address the churn issue, we should review our customer service strategy, focusing on factors such as average service calls, which range from 1.52 to 1.6 across cohorts, to identify potential areas for improvement.

Churn by tenure cohort and service-call bucket

The service-call cliff holds in every cohort. Tenure itself does not predict churn.



Tenure is the ordered axis in this project because the dataset has no date column. It is useful for structure, not as a churn driver.

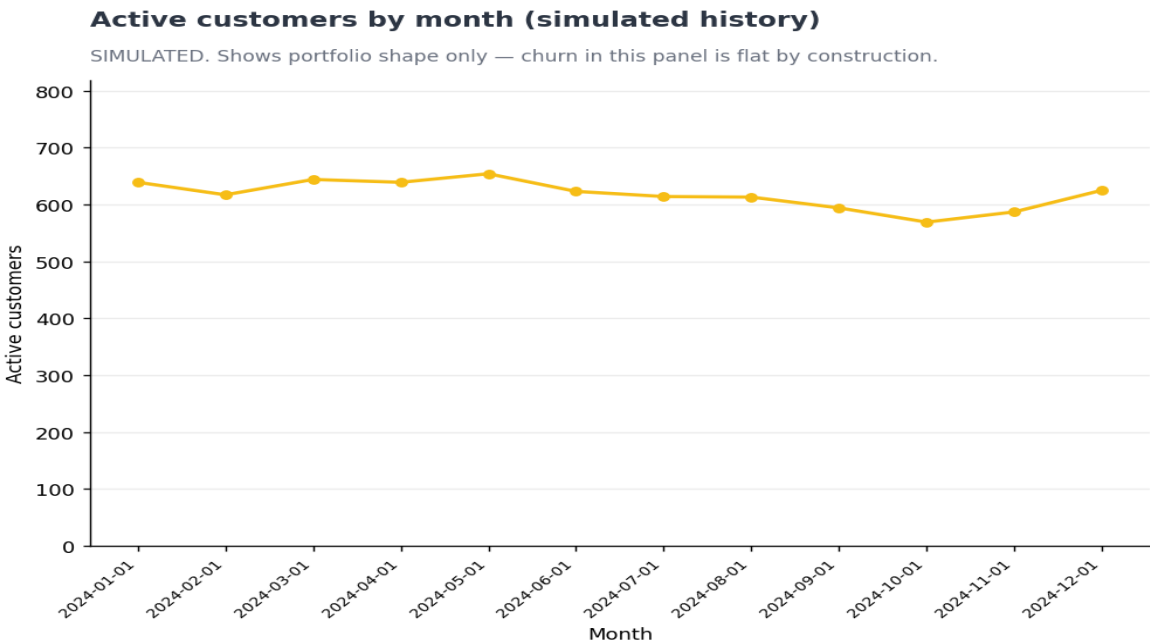
AI explanation. The pattern shows that within each tenure cohort, the churn rate is similar for customers with 0-3 service calls, but jumps sharply for those with 4 or more service calls. For example, in the New cohort, the churn rate is 10.79% for customers with 0-3 service calls, but 51.72% for those with 4 or more. This suggests that the number of service calls is a key driver of churn, rather than the length of time a customer has been with the company. To address this, the company should review its customer support processes to identify opportunities to reduce the number of service calls, such as implementing more effective issue resolution or proactive outreach to customers.

Portfolio structure over time‡

SIMULATED HISTORY. This section is derived from generated monthly snapshots, not observed history. It shows portfolio structure only. Churn in this panel is flat by construction and cannot support a trend claim or a forecast.

12	569-654	\$8,529-\$9,954
Months shown	Active base	Monthly revenue

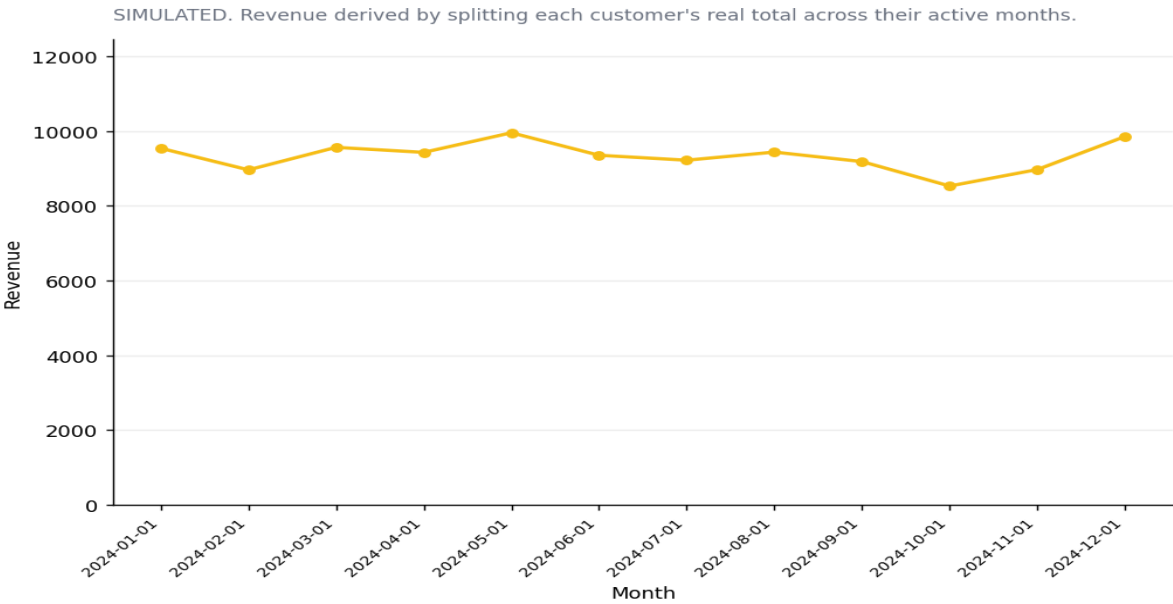
The portfolio's structure is characterized by an active customer base ranging from 569 to 654 customers, with a revenue range of \$8529.49 to \$9954.1. The average revenue per user (ARPU) varies from \$14.53 to \$15.76. It's essential to note that this history is simulated, and any observations are based on a single snapshot. The data does not support temporal comparisons, and any variation in metrics should not be interpreted as a trend. The portfolio's revenue and active customer base can be compared across different segments, but not across time.



Generated from the snapshot, not observed. Included to show portfolio shape; it contains no information the snapshot did not already hold.

AI explanation. This history is simulated, and the active base appears to be relatively stable, with active customers ranging from 569 to 654. A possible explanation for this stability is that entries and exits are roughly in balance across the period shown. The active customer count is highest at 654 and lowest at 569. To better understand the portfolio structure, review the customer acquisition and retention strategies to identify opportunities for growth and optimization.

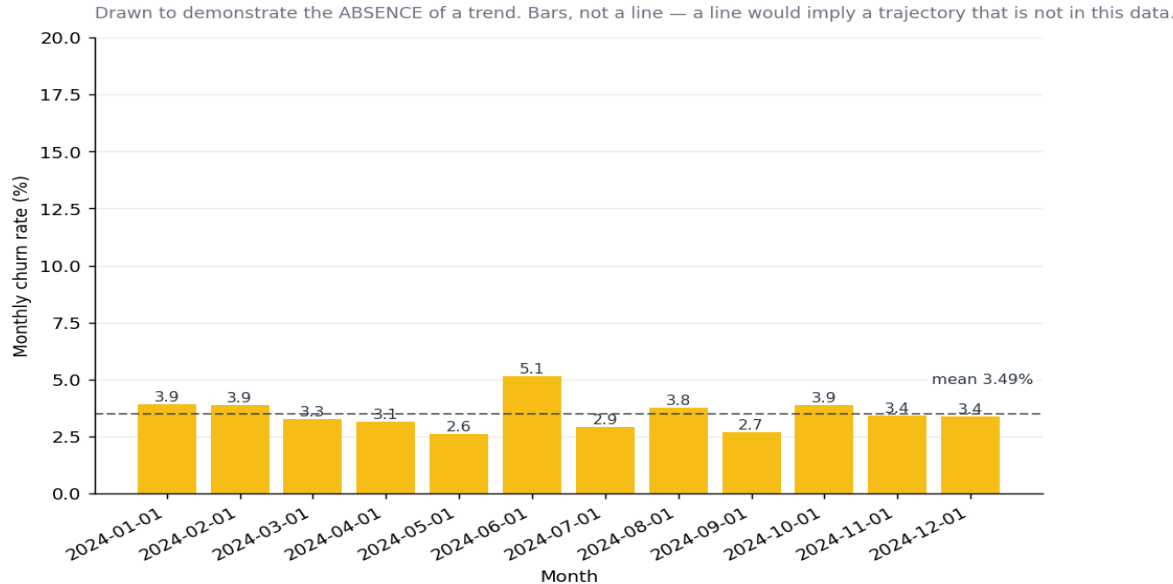
Monthly revenue (simulated history)



Each customer's real total charge was distributed across their active months. Monthly totals sum back to the real portfolio revenue.

AI explanation. The pattern of monthly revenue is relatively stable, with total revenue ranging from 9183.23 to 9954.1. This stability suggests that the simulated history is not driven by significant fluctuations in customer behavior. The data does not support temporal comparison, as this history is simulated, not observed. Given the stability of monthly revenue, a specific action to take would be to review the revenue composition, such as the day share percentage, which remains relatively consistent at around 51.3 to 51.7 percent, to identify opportunities for optimization.

Monthly churn is FLAT by construction (simulated)



Included to demonstrate the absence of a trend, not to show one. Churn timing in this panel was generated; a different random seed would produce a different arrangement of the same flat distribution. Real churn movement over time cannot be measured from a single snapshot.

AI explanation. The pattern in this chart is essentially flat, with all months sitting close to the mean of 3.49%. This history is simulated, and the variation between months is sampling noise from the generator, not customer behaviour. The absence of a pattern is itself informative, suggesting that the generator did not encode a trend that the source data could not support. A possible explanation for this is that the simulation was designed to demonstrate the absence of a trend. Based on this, a concrete step would be to review the simulation parameters to ensure they accurately reflect the underlying data, using the mean churn rate of 3.49% as a reference point.

What this report cannot tell you

- **NO TIME DIMENSION.** The source data is a single snapshot with no dates. Nothing here describes how any metric changed over time, and no figure in this report is a forecast.
- **SIMULATED HISTORY SHOWS STRUCTURE, NOT TRENDS.** The monthly panel in this report is generated from the snapshot, so it contains no information the snapshot did not already hold. Churn in it is deliberately flat: a trend there would reflect the generator's random seed, not customer behaviour. It is included to show portfolio shape and tenure lifecycle, both of which are legitimate uses of derived monthly data.
- **TENURE IS NOT A DRIVER.** Churn is flat across tenure cohorts (13.1%-15.3%). This was expected to matter and does not - a genuine finding, and a reason not to target by tenure.
- **STATE-LEVEL FIGURES ARE NOISY.** 51 states average about 65 customers each, so a handful of churners moves a state several percentage points. Geography is the weakest dimension here.
- **PROJECTED VALUES REST ON ASSUMPTIONS.** Save rate, contact cost and acquisition cost are industry-typical placeholders, not measurements: the dataset contains no cost data. A save rate of 30% is assumed, with a sensitivity band of 15%-45%.
- **REVENUE IS PERIOD CHARGES.** Not monthly recurring revenue and not annualised.
- **CHURN RISK IS NOT PREDICTED HERE.** Segments describe which customers resemble those who already churned. Scoring individual customers is a separate modelling problem.

Provenance

Every observed figure traces to the source data. Segment totals reconcile to the portfolio: customers match (3,333), revenue matches (\$198,146.03).

All narrative text passed automated validation: every figure quoted was checked against the data that produced it, and no claim about change over time was permitted.

Warning: the AI explanation beneath these charts failed validation and is marked in red: revenue_at_risk.